

A robust numerical scheme for a time-fractional Black-Scholes partial differential equation describing stock exchange dynamics[☆]

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ABSTRACT

Empirical evidence suggest that fractional stochastic based models are well suited for modelling systems and phenomena exhibiting memory and hereditary properties. Assuming that the stock market exhibits some unexplained memory structures, described by a non-random fractional stochastic process governed under a standard Brownian motion, we derive a time-fractional Black-Scholes (tfBS) partial differential equation for pricing option contracts on such stocks. We further propose a corresponding robust numerical method which is based on the extension of a Crank Nicholson finite difference method for solving tfBS-PDEs. Through rigorous theoretical analysis, we established that the method is unconditionally stable and convergent up to order $\mathcal{O}(k^2 + h^2)$. Two numerical examples are presented using realistic market parameters. Our results confirm theoretical observations and general consensus in literature that, stock market dynamics are of a power law nature and follow heavy tailed distributions with memory.

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1. Introduction

The classical asset pricing model originally proposed by Black & Scholes in their famous paper [3] suffers from a few drawbacks. These drawbacks are naturally emanating from its strict design assumptions. Among them is the assumption that financial assets' returns follow a normal distribution. In the earlier days, the classical Black-Scholes approach was regarded as a very efficient and robust approach, and as such, it received enormous applications in pricing a number of financial assets and contracts.

Amongst other financial derivatives, the model has enjoyed a considerable attention in valuing financial options, in particular, those of the American style or European ones. The main idea of the classical Black-Scholes approach lies in the construction of a portfolio consisting of a risk-free asset, e.g., a bond, coupled with a risky asset, e.g., a stock. The portfolio is constructed in such a way that it does not allow for arbitrage opportunities.

Besides the vast applications of the above approach in derivatives pricing, change in market dynamics due to ill-informed splash

trading, for example, has led to formations of a number of speculative bubbles and volatility smiles. Often when these speculative bubbles burst, unfavourable economic recessions and crises are inevitable. Comprehensive accounts on the role of classical Black-Scholes approach in speculative bubbles and volatility smiles formation, and works on economic crises in general can be found in [1,4,11] and references therein. The existence of memory or hereditary structures in a number of financial systems has long been documented in financial literature, see for example, [6,17,18,22–24].

In reality and also contrary to what the efficient market hypothesis (EMH) may suggest, unusual structures and patterns observed in a number of financial markets data can strongly be attributed to the following three factors, (i) markets discount everything, (ii) asset prices moves in trends and patterns forming structures depicting investors' psychology, and (iii) these repeated market structures may persist and are most likely to re-occur more often than expected (history mostly always repeat itself).

According to Panas [18], local derivative based mathematical models can only capture localized information about change in the price at a particular point and time. As such, these models may not be appropriate for modeling dynamics of markets depicting unusual structures such as jumps and repeated patterns. Panas [18], indicated that memory effects in financial data occurs in two forms: memory due to the noise and that due to the trend. He

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further mentioned that incorporating a fractional Brownian motion as the governing process of the underlying dynamics can only help to capture the noise memory effects and not the trend memory effects.

Until recently, when the fractal nature of financial markets were discovered, memory effects were long measured using auto-correlation functions. Based on a series of his publications on the theory of fractal geometry, Mandelbrot [14] has certainly revolutionising the world of science. With the fields of science, finance and economics being some of his most subtle areas of interests, led him onto the path of discovering the fractal geometry of financial markets (fractal market hypothesis-FMH). The work by Mandelbrot and Cioczek-Georges [15,16] put them at the forefront of pioneers in the area of application of the fractional models to financial modeling and literature. They suggested an approach where they analogously replaced the standard Brownian motion in the classical Black-Scholes model with a fractional Brownian motion (fBM) characterised by a Hurst parameter $H \in [0, 1]$. Unlike in the standard case, the hurst parameter H in the fBM helps in explaining the effects of memory. Some of the recent works on the similar approach are Jumarie [7], Liang [13] and Wei-Gou [22] just to mention but a few.

Motivated by the ideal that fractional calculus is a powerful tool for explaining effects of memory observed in a number of physical systems and phenomena, a growing number of research work on fractional Black-Scholes models has been published. Due to the perceived theoretical difficulties involved in defining solutions to fractional Brownian motions based models, Jumarie [10] suggested an alternative, which is to consider a non-random fractional stochastic dynamics subject to the standard Brownian motion instead of direct replacement of the governing process with a fractional Brownian motion. According to Jumarie [10], this approach was only possible due to the discovery of a well defined fractional calculus framework, namely, fractional Taylor series theory, which was initially introduced by Osler [12] and later refined by Jumarie in [7].

Models resulting from the above mentioned approach are generically referred to as fractional Black-Scholes partial differential equations (fBS-PDEs). So far, there are only two main classes of fBS-PDEs, the time-fractional Black-Scholes PDEs (tfBS-PDEs) and the time-and-space fractional Black-Scholes PDEs (tsfBS-PDEs). The pioneering ideas in the derivation of these PDEs and other subsequent fractional theories therein can strongly be attributed to a series of work by Jumarie in [7,9,10]. Numerous other authors built onto this idea either by suggesting or deriving semi-analytic solutions mostly via fractional Laplace transform or by suggesting numerical solutions for solving these PDEs.

In recent work, a q -homotopy analysis transform method (q -HATM) was proposed in [20]. This technique is an amalgamation of the Laplace transform technique with q -homotopy analysis scheme and fractional derivative defined using the Atangana-Baleanu operator. In Wei et al. [5] a numerical solution of a fractional calculus based model describing the deadly disease in pregnant women was proposed. This method was also used in [20] to solve a system of three fractional differential equations describing the Lakes pollution. In [21] the same approach was used for solving a fractional Schistosomiasis disease model. Their results also indicate that the method is very effective and easy to apply when examining multi-dimensional differential equations of fractional order arising in biological phenomena.

Though semi-analytic and analytic solutions such as those obtained using the Laplace transform and have been suggested, their applications and interpretations on pricing financial derivatives still remain a subject for further investigations. Therefore, this paper suggest a robust finite difference based numerical scheme for solving tfBS-PDEs in pricing European stock options. The ideas and

techniques presented herein can be extended for solving tsfBS-PDEs. The method under consideration is based on the extension of the Crank-Nicholson (CN) method for solving tfBS-PDEs. To the best of our knowledge, this approach has not yet been explored for solving this class of problems.

For the rest of this paper is organized as follow: Section 2 presents some fundamental ideas on the concept of fractional derivatives with specific reference on the three common definitions of fractional derivatives, the Caputo, Reimann-Liouville and the Jumarie-revised Reiman-Liouville derivative. This section also presents a brief discussion on the derivation of the tfBS-PDE in relation to pricing European options on continuous dividend paying stocks. In Section 3, we present the detailed construction of the numerical scheme. A comprehensive theoretical analysis of the method in terms of convergence and stability is presented in Section 4. Two practical examples on the use of the approach for pricing European put options on stocks are presented in Section 5. Lastly in Section 6 we present some concluding remarks and scope for future research.

2. The time-fractional Black-Scholes PDE

This section present an overview of preliminary material on the subject of fractional differentiation while specifying the involved tfBS-model and its derivation.

2.1. Mathematical preliminaries on fractional differentiation

The most commonly used definitions of fractional derivatives in recent literature are defined in the Caputo, Reimann-Liouville and the Jumarie-revised Reimann-Liouville sense. Though there exists numerous other definitions, the trio posses specific desirable properties appropriate to modelling financial data.

To develop the basis of our tfBS-PDE and also touch base with the concept of fractional calculus on application to financial modelling and analysis, we will briefly revisit the above three definitions. We will specifically focus on their mathematical formulations, their merits and de-merits.

Definition 2.1. Caputo Derivative. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous, but not necessarily a differentiable function. The Caputo fractional derivative of order α as defined in Jumarie ([8]) is defined as

$$D_c^\alpha f(t) = \frac{1}{\Gamma(\eta - \alpha)} \int_0^t \frac{1}{(t - \tau)^{\alpha - \eta + 1}} \frac{d^\eta f(\tau)}{d\tau^\eta} d\tau, \quad \eta - 1 < \alpha \leq \eta, \quad \eta \in \mathbb{N}. \quad (2.1)$$

The Caputo fractional derivative of a non-differentiable function has a kernel at the origin and its derivative of a constant function is not zero (see for example [2,10]) and references therein. However, according to [19] one of the main advantage of this definition is the fact that it allows for incorporation of traditional initial and boundary conditions into the formulation of the problem which provides a framework that is well consistent with the classical approach.

Definition 2.2. Reimann-Liouville Derivative. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function, but not necessarily differentiable. Then the Reimann-Liouville fractional derivative of order α is given by (see [2,8,9] among others)

$$D_{RL}^\alpha f(t) = \frac{1}{\Gamma(\eta - \alpha)} \frac{d^\eta}{dt^\eta} \int_0^t \frac{f(\tau)}{(t - \tau)^{\alpha - \eta + 1}} d\tau, \quad \eta - 1 < \alpha \leq \eta, \quad \eta \in \mathbb{N}. \quad (2.2)$$

Readers may note that like the Caputo derivative, the Riemann-Liouville derivative of a constant is non-zero, and the derivative of

any function which is constant at the origin, for example the exponential or Mittag-Leffler function, have singularities at the origins. Due to the shortcomings in the Caputo and Riemann-Liouville definitions, their applications in modelling a number of non-linear real-life problems have been limited [2].

A modified definition by Jumarie [8] along with the fractional (Generalized) Taylor series theory found in [8,12] addresses the issues of singularities about the origin, as well the non-zero constraints of constant functions.

Definition 2.3. Jumarie (revised Riemann-Liouville) Derivative. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function, but not necessarily a differentiable, and suppose that $f(t)$ is

- (i) a constant K , then its Jumarie fractional derivative of order α is defined by

$$D_f^\alpha f(t) = \begin{cases} \frac{K}{\Gamma(\eta-\alpha)} t^{-\alpha+1-\eta}, & \alpha \leq \eta - 1, \eta \in \mathbb{N}, \\ 0, & \alpha > \eta - 1, \eta \in \mathbb{N}, \end{cases} \quad (2.3)$$

- (ii) not a constant, then

$$D_f^\alpha f(t) = \frac{1}{\Gamma(\eta-\alpha)} \frac{d^\eta}{dt^\eta} \int_0^t \frac{\{f(\tau) - f(0)\}}{(t-\tau)^\alpha} d\tau, \quad \eta - 1 < \alpha \leq \eta, \eta \in \mathbb{N}. \quad (2.4)$$

The Jumarie definition in Eqn 2.4 takes into account the existence of a fractional derivative at the origin, and the existence of fractional derivative of a constant, hence aligning the definition consistently well with local derivative based differential calculus.

In what follows, we briefly present some key ideas involved in the derivation of the involved tfBS-PDE.

2.2. Model specification

Suppose the stock price S follows the following non-random fractional stochastic process presented in a fractional Maruyama representation $dB_\alpha(t) = \omega(t)(dt)^\alpha/2$, which leads to

$$dS = (r - \delta)Sdt + \sigma S\omega(t)(dt)^\alpha/2, \quad 0 < \alpha \leq 1, \quad (2.5)$$

where $B_\alpha(t)$ is a fractal process governed by a Gaussian white noise $\omega(t)$, with σ^2 representing the volatility while r and δ represent the risk-free interest rate and the continuous dividend yield, respectively. The fractional dynamic equation (2.5) generalize the standard Brownian motion $B(t)$ given by $dB(t) = \omega(t)dt^{1/2}$ with $dB_\alpha(t) = \omega(t)(dt)^\alpha/2$ where $0 < \alpha \leq 1$. Note that when $\alpha = 1$, equation (2.5) is equivalent to a geometric Brownian motion.

It should be noted that unlike in the standard Brownian motion, the non-Gaussian fractional process (2.5) does not make any prior assumption about the underlying distribution of the stock price. However, it does make inferences on how the market is scaling with respect to time. This feature is very important in modelling market cycles in forms of repeated patterns.

From equation (2.5) one can derive two families of fractional Black-Scholes PDEs, namely; tfBS-PDEs as well as tsfBS PDEs. Fundamental to the derivation of the involved tfBS-PDE are the following three key conversion formulae which are useful to convert integer and fractional derivatives. According to [10] the following identities are well consistent with the fractional (Generalized) Taylor series expansion.

$$d^\alpha t = \frac{1}{\Gamma(2-\alpha)} t^{1-\alpha} (dt)^\alpha, \quad 0 < \alpha \leq 1, \quad (2.6)$$

$$d^\alpha S = \Gamma(1+\alpha) dS, \quad 0 < \alpha \leq 1, \quad (2.7)$$

and

$$\frac{d^\alpha S}{(dS)^\alpha} = \frac{1}{\Gamma(2-\alpha)} S^{1-\alpha}, \quad 0 < \alpha \leq 1. \quad (2.8)$$

The fractional derivative herein is defined in the Caputo sense as defined in (definition 2.1) when $\eta = 1$. Combination of the two identities (2.7) and (2.8) and aided by Eqn 2.6 results in the following derivative conversion formula

$$dS = \frac{S^{(1-\alpha)}}{\Gamma(1+\alpha)\Gamma(2-\alpha)} (dS)^\alpha, \quad 0 < \alpha \leq 1. \quad (2.9)$$

Let $V = V(S, t)$ be the value of a European option, such that, V is sufficiently smooth with respect to S and t , and that, its time-fractional derivative of order α ($0 < \alpha \leq 1$) exists. Then, combining the interest evolution equation of typical bank savings account in Eqn 2.10

$$dV = rVdt, \quad (2.10)$$

with the conversion formula (2.9) results in the following fractional interest rate equation

$$d^\alpha V = \frac{rV}{\Gamma(2-\alpha)} t^{1-\alpha} (dt)^\alpha. \quad (2.11)$$

Since V is sufficiently smooth with respect to S and fractionally differentiable in time up to order α ($0 < \alpha \leq 1$), then, the fractional Taylor series expansion of V up to some remaining error terms yields

$$dV = \frac{1}{\Gamma(1+\alpha)} \frac{\partial^\alpha V}{\partial t^\alpha} (dt)^\alpha + \frac{\partial V}{\partial S} dS + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} (dS)^2. \quad (2.12)$$

Combining (2.11) with the fractional Itô's lemma applied to equation (2.5), coupled with some algebraic manipulations, we obtain the tfBS-PDE

$$\frac{\partial^\alpha V}{\partial t^\alpha} = \left(rV - qS \frac{\partial V}{\partial S} \right) \frac{t^{1-\alpha}}{\Gamma(2-\alpha)} - \frac{\Gamma(1+\alpha)}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}, \quad q = r - \delta; \quad 0 < \alpha \leq 1, \quad (2.13)$$

subject to the following initial and boundary conditions for a European put option

$$\left. \begin{aligned} V(S, 0) &= \max(K - S, 0), \\ V(0, t) &= Ke^{-r(T-t)}, \\ \lim_{S \rightarrow \infty} V(S, t) &= 0, \end{aligned} \right\} \quad (2.14)$$

with K the strike price and T the maturity time. In the next section we present the details and construction of the numerical method. A similar model to (2.13) was also derived in [9] describing stock exchange dynamics with fractional exponential growth.

3. Model discretization and numerical scheme

This section present the discretization of derivative terms in the tfBS-PDE (2.13) as well as its full final scheme subject to initial and boundary conditions in (2.14).

3.1. Model discretization

Let L and N be positive integers and define $h = S_{max}/L$ and $k = T/N$ as the space and time step-sizes, respectively. Define $S_l = lh$; $l = 0, 1, 2, \dots, L$ and $t_n = nk$; $n = 0, 1, 2, \dots, N$, such that $S_l \in [S_{min}, S_{max}]$ and $t_n \in [0, T]$. Furthermore, define $V_l^{n+1} = V(S_l, t_{n+1})$ as the solution at the grid point $(S_l, t_{n+1}) = (lh, (n+1)k)$.

3.1.1. Temporal discretization

Since the Caputo definition allows for incorporation of traditional initial and boundary conditions in the problem formulation, and also due to the fact that the value of a European option $V(S, t)$ is a time differentiable function, we define the time-fractional derivative in (2.13) using Definition 2.1. If we let $\eta = 1$,

the time-fractional derivative in (2.13) can be discretized as follows. We have

$$\begin{aligned}
 \frac{\partial^\alpha V(S_l, t_{n+1})}{\partial t^\alpha} &= \frac{1}{\Gamma(1-\alpha)} \int_0^{t_{n+1}} \frac{\partial V(S_l, \tau)}{\partial \tau} (t_{n+1} - \tau)^{-\alpha} d\tau, \\
 &\leq \frac{1}{\Gamma(1-\alpha)} \sum_{j=1}^{n+1} \int_{j\tau}^{(j+1)\tau} \left(\frac{V_l^{j+1} - V_l^{j-1}}{2k} + \mathcal{O}(k^2) \right) ((n+1)k - \tau)^{-\alpha} d\tau, \\
 &= \frac{1}{\Gamma(1-\alpha)} \frac{1}{1-\alpha} \sum_{j=1}^{n+1} \left(\frac{V_l^{j+1} - V_l^{j-1}}{2k} + \mathcal{O}(k^2) \right) \\
 &\quad [(n-j)^{1-\alpha} - (n-j+1)^{1-\alpha}] k^{1-\alpha}, \\
 &= \frac{1}{2\Gamma(2-\alpha)} \frac{1}{k^\alpha} \sum_{j=1}^{n+1} [V_l^{j+1} - V_l^{j-1}] [(n-j)^{1-\alpha} - (n-j+1)^{1-\alpha}] \\
 &\quad + \frac{1}{\Gamma(2-\alpha)} \sum_{j=1}^{n+1} [(n-j)^{1-\alpha} - (n-j+1)^{1-\alpha}] \mathcal{O}(k^2) k^{1-\alpha}, \\
 &= \frac{1}{2\Gamma(2-\alpha)} \frac{1}{k^\alpha} \sum_{j=1}^{n+1} (V_l^{j+1} - V_l^{j-1}) [(n-j)^{1-\alpha} - (n-j+1)^{1-\alpha}] \\
 &\quad + \frac{1}{\Gamma(2-\alpha)} \sum_{j=1}^{n+1} [(n-j)^{1-\alpha} - (n-j+1)^{1-\alpha}] \mathcal{O}(k^{3-\alpha}). \tag{3.1}
 \end{aligned}$$

Shifting the indices in (3.1) from j to $n-j$, we obtain

$$\begin{aligned}
 \frac{\partial^\alpha V(S_l, t_n)}{\partial t^\alpha} &= \frac{1}{2\Gamma(2-\alpha)} \frac{1}{k^\alpha} \sum_{j=0}^n (V_l^{n-j+1} - V_l^{n-j-1}) [j^{1-\alpha} - (j+1)^{1-\alpha}] \\
 &\quad + \frac{1}{2\Gamma(2-\alpha)} \sum_{j=0}^n [j^{1-\alpha} - (j+1)^{1-\alpha}] \mathcal{O}(k^{3-\alpha}). \tag{3.2}
 \end{aligned}$$

Define

$$\rho_\alpha := -\frac{1}{2\Gamma(2-\alpha)} \frac{1}{k^\alpha}, \tag{3.3}$$

and

$$\beta_j := (j+1)^{1-\alpha} - j^{1-\alpha}, \quad j = 0, \dots, n, \tag{3.4}$$

where, $1 = \beta_0 > \beta_1 > \beta_2 > \dots > 0$ and $\lim_{n \rightarrow \infty} \sum_{j=0}^n \beta_j = n^{1-\alpha}$. Substituting ρ_α and β_j into (3.2) we obtain

$$\begin{aligned}
 \frac{\partial^\alpha V(S_l, t_{n+1})}{\partial t^\alpha} &= \rho_\alpha \sum_{j=0}^n \beta_j (V_l^{n-j+1} - V_l^{n-j-1}) - \frac{1}{\Gamma(2-\alpha)} \sum_{j=0}^n \beta_j \mathcal{O}(k^{3-\alpha}), \\
 &\leq \rho_\alpha \sum_{j=0}^n \beta_j (V_l^{n-j+1} - V_l^{n-j-1}) - \frac{1}{\Gamma(2-\alpha)} n^{1-\alpha} \mathcal{O}(k^{3-\alpha}), \\
 &= \rho_\alpha \sum_{j=0}^n \beta_j (V_l^{n-j+1} - V_l^{n-j-1}) - \frac{1}{\Gamma(2-\alpha)} \left(\frac{t_n}{k} \right)^{1-\alpha} \mathcal{O}(k^{3-\alpha}), \\
 &= \rho_\alpha \sum_{j=0}^n \beta_j (V_l^{n-j+1} - V_l^{n-j-1}) - \frac{t_n^{1-\alpha}}{2\Gamma(2-\alpha)} k^2. \tag{3.5}
 \end{aligned}$$

The time derivative in (2.13) is therefore approximated by

$$\frac{\partial^\alpha V(S_l, t_{n+1})}{\partial t^\alpha} = \rho_\alpha \sum_{j=0}^n \beta_j (V_l^{n-j+1} - V_l^{n-j-1}) + \mathcal{O}(k^2). \tag{3.6}$$

3.1.2. Spatial discretization

The first and second spatial derivatives in (2.13) are approximated respectively by

$$\frac{\partial V(S_l, t_{n+1})}{\partial S} = \frac{V_{l+1}^{n+1} - V_{l-1}^{n+1}}{2h} + \mathcal{O}(h^2), \tag{3.7}$$

and

$$\frac{\partial^2 V(S_l, t_{n+1})}{\partial S^2} = \frac{V_{l+1}^{n+1} - 2V_l^{n+1} + V_{l-1}^{n+1} + V_{l+1}^n - 2V_l^n + V_{l-1}^n}{2h^2} + \mathcal{O}(h^2). \tag{3.8}$$

3.2. The full scheme

Substituting (3.6), (3.7) and (3.8) into (2.13), we obtain for $n \geq 0$:

$$\rho_\alpha \sum_{j=0}^n \beta_j (V_l^{n-j+1} - V_l^{n-j-1}) = \frac{t^{1-\alpha}}{\Gamma(2-\alpha)} (rV_l^{n+1} - ql(V_{l+1}^{n+1} - V_{l-1}^{n+1})) - \frac{\Gamma(1+\alpha)\sigma^2 l^2 (V_{l+1}^{n+1} - 2V_l^{n+1} + V_{l-1}^{n+1} + V_{l+1}^n - 2V_l^n + V_{l-1}^n)}{4}. \quad (3.9)$$

The above scheme is of the Crank-Nicholson type, which can be broken down into two separate distinct cases, i.e., for when $n = 0$ and when $n \geq 1$.

Setting $n = 0$, and equating the non-fictitious point to zero we obtain the following system of equations

$$a_1 V_{l-1}^1 + b_1 V_l^1 + c_1 V_{l+1}^1 = a_0 V_{l-1}^0 + b_0 V_l^0 + c_0 V_{l+1}^0, \quad (3.10)$$

where

$$\begin{aligned} a_1 &= \sigma^2 l^2 \frac{\Gamma(1+\alpha)}{4} - ql \frac{t^{1-\alpha}}{\Gamma(2-\alpha)}, \\ b_1 &= \rho_\alpha \beta_0 - r \frac{t^{1-\alpha}}{\Gamma(2-\alpha)} - \sigma^2 l^2 \frac{\Gamma(1+\alpha)}{2}, \\ c_1 &= \sigma^2 l^2 \frac{\Gamma(1+\alpha)}{4} + ql \frac{t^{1-\alpha}}{\Gamma(2-\alpha)}, \end{aligned}$$

and

$$\begin{aligned} a_0 &= -\sigma^2 l^2 \frac{\Gamma(1+\alpha)}{4}, \\ b_0 &= \rho_\alpha \beta_0 + \sigma^2 l^2 \frac{\Gamma(1+\alpha)}{2}, \\ c_0 &= -\sigma^2 l^2 \frac{\Gamma(1+\alpha)}{4}. \end{aligned}$$

For $n \geq 1$:

$$\begin{aligned} a_{n+1} V_{l-1}^{n+1} + b_{n+1} V_l^{n+1} + c_{n+1} V_{l+1}^{n+1} &= a_n V_{l-1}^n + b_n V_l^n + c_n V_{l+1}^n \\ &+ \rho_\alpha \sum_{j=1}^n \varphi_j V_l^{n-j+1} + \rho_\alpha \beta_{n+1} V_l^0 \end{aligned} \quad (3.11)$$

where

$$\begin{aligned} a_{n+1} &= \sigma^2 l^2 \frac{\Gamma(1+\alpha)}{4} - ql \frac{t^{1-\alpha}}{\Gamma(2-\alpha)}, \\ b_{n+1} &= \rho_\alpha \beta_1 - r \frac{t^{1-\alpha}}{\Gamma(2-\alpha)} - \sigma^2 l^2 \frac{\Gamma(1+\alpha)}{2}, \\ c_{n+1} &= \sigma^2 l^2 \frac{\Gamma(1+\alpha)}{4} + ql \frac{t^{1-\alpha}}{\Gamma(2-\alpha)}, \end{aligned}$$

and

$$\left. \begin{aligned} a_n &= -\sigma^2 l^2 \frac{\Gamma(1+\alpha)}{4}, \\ b_n &= \rho_\alpha \beta_1 + \sigma^2 l^2 \frac{\Gamma(1+\alpha)}{2}, \\ c_n &= -\sigma^2 l^2 \frac{\Gamma(1+\alpha)}{4}, \end{aligned} \right\} \quad (3.12)$$

$$\varphi_j = \beta_j - \beta_{j+1}; \quad j = 0, 1, 2, \dots, n,$$

which leads to the following general matrix representation

$$\mathbf{A}\mathbf{U}^{n+1} = \mathbf{B}\mathbf{U}^n + \sum_{j=1}^n \varphi_j \mathbf{U}^{n-j+1} + \mathbf{C}^n, \quad \text{for all } n \geq 1, \quad (3.13)$$

where

$$\mathbf{A} = \begin{pmatrix} a_{n+1} & b_{n+1} & c_{n+1} & 0 & \cdots & \cdots & 0 \\ 0 & a_{n+1} & b_{n+1} & c_{n+1} & & & \vdots \\ \vdots & & \ddots & \ddots & \ddots & & \vdots \\ \vdots & & & a_{n+1} & b_{n+1} & c_{n+1} & 0 \\ 0 & 0 & \cdots & 0 & a_{n+1} & b_{n+1} & c_{n+1} \end{pmatrix},$$

$$\mathbf{B} = \begin{pmatrix} a_n & b_n & c_n & 0 & \cdots & \cdots & 0 \\ 0 & a_n & b_n & c_n & & & \vdots \\ \vdots & & \ddots & \ddots & \ddots & & \vdots \\ \vdots & & & a_n & b_n & c_n & 0 \\ 0 & 0 & \cdots & 0 & a_n & b_n & c_n \end{pmatrix}, \quad (3.14)$$

$$\mathbf{U}^n = \begin{pmatrix} V_0^n \\ V_1^n \\ \vdots \\ V_{L-1}^n \\ V_L^n \end{pmatrix},$$

and

$$\mathbf{C}^n = \rho_\alpha \beta_n \mathbf{U}_l^0, \text{ for all } n \geq 1.$$

Remark 3.1. The following observations are trivial to show

$$\left. \begin{aligned} 1 = \beta_0 > \beta_1 > \cdots \rightarrow 0, \\ \varphi_0 = 1 - \beta_1, \\ \sum_{j=0}^n \varphi_j = 1 - \beta_{n+1}, \\ \sum_{j=0}^\infty \varphi_j = 1 > 2^{1-\alpha} - (2^{1-\alpha} - 1^{1-\alpha}) = 1 = \varphi_0 > \varphi_1 > \cdots \rightarrow 0. \end{aligned} \right\} \quad (3.15)$$

4. Analysis of the numerical method

This section presents the stability and convergence analysis of the scheme in (3.10)-(3.11). The analysis is based on the Fourier series analysis for fractional PDEs.

4.1. Stability analysis of the numerical method

Theorem 4.1. The scheme (3.10)-(3.11) along with (2.14) is unconditionally stable.

Proof. Let $v_l^n, l = 0, 1, 2, \dots, L; n = 0, 1, 2, \dots, N$; be an approximate solution to the difference equations (3.10)-(3.11), and define $\epsilon_l^n = V_l^n - v_l^n$ as the truncation error, such that $\epsilon_0^n = \epsilon_L^n = 0$ for all n . Since the approximate solution satisfy the two equations (3.10)-(3.11), then, substituting ϵ_l^n into (3.10)-(3.11) yield

$$a_{1l}\epsilon_{l-1}^1 + b_{1l}\epsilon_l^1 + c_{1l}\epsilon_{l+1}^1 = a_{0l}\epsilon_{l-1}^0 + b_{0l}\epsilon_l^0 + c_{0l}\epsilon_{l+1}^0; \quad n = 0, \quad (4.1)$$

and

$$a_{n+1l}\epsilon_{l-1}^{n+1} + b_{n+1l}\epsilon_l^{n+1} + c_{n+1l}\epsilon_{l+1}^{n+1} = a_{nl}\epsilon_{l-1}^n + b_{nl}\epsilon_l^n + c_{nl}\epsilon_{l+1}^n + \sum_{j=0}^n \varphi_j \epsilon_l^{n-j+1} + \beta_{n+1}\epsilon_l^0; \quad (4.2)$$

$$n \geq 1.$$

Consider the function

$$\epsilon^n(S) = \begin{cases} \epsilon_l^n, & \text{when } S_l - \frac{h}{2} < S \leq S_l + \frac{h}{2}, \quad l = 1, 2, \dots, L-1, \\ 0, & \text{when } 0 \leq S < \frac{h}{2} \text{ or } S_{\max} - \frac{h}{2} < S \leq S_{\max} + \frac{h}{2}. \end{cases} \quad (4.3)$$

The $\epsilon^n(S)$ can be represented by the Fourier series

$$\epsilon^n(S) = \sum_{j=1}^{\infty} \varrho_n(j) e^{i2\pi jS/S_{\max}}; \quad n = 0, 1, 2, \dots, N, \quad (4.4)$$

for

$$\varrho_n(j) = \frac{1}{S_{\max}} \int_0^{S_{\max}} \epsilon^n(S) e^{-i2\pi jS/S_{\max}} dS; \quad n = 0, 1, 2, \dots, N. \quad (4.5)$$

Let $\epsilon^n = (\epsilon_1^n, \epsilon_2^n, \dots, \epsilon_{L-1}^n)^T$ and define it's norm by

$$\|\epsilon^n\|_2 = \left(\sum_{l=1}^{L-1} h |\epsilon_l^n|^2 \right)^{1/2} = \left(\int_0^{S_{\max}} |\epsilon^n(S)|^2 dS \right)^{1/2}. \quad (4.6)$$

Then, using the Parseval equality

$$\int_0^{S_{\max}} |\epsilon^n(S)|^2 dS = \sum_{j=-\infty}^{\infty} |\varrho_n(j)|^2, \quad (4.7)$$

we obtain

$$\|\epsilon^n\|_2^2 = \int_0^{S_{\max}} |\epsilon^n(S)|^2 dS = \sum_{j=-\infty}^{\infty} |\varrho_n(j)|^2. \quad (4.8)$$

Therefore, the solution to (4.1) and (4.2) takes the following form

$$\epsilon^n = \varrho_n e^{i\beta h}, \quad (4.9)$$

where $\beta = 2\pi j/S_{\max}$ and $i = \sqrt{-1}$. Substituting ϵ^n into (4.1) and (4.2), we obtain

$$(a_1 e^{-\beta h} + c_1 e^{\beta h} + b_1) \varrho_1 = (a_0 e^{-\beta h} + c_0 e^{\beta h} + b_0) \varrho_0; \text{ for } n = 0, \quad (4.10)$$

whereas for $n \geq 1$, we obtain

$$(a_{n+1} e^{-\beta h} + c_{n+1} e^{\beta h} + b_{n+1}) \varrho_{n+1} = (a_n e^{-\beta h} + c_n e^{\beta h} + b_n) \varrho_n + \sum_{j=1}^n \varphi_j \varrho_{n+1-j} + \beta_{n+1} \varrho_0, \quad (4.11)$$

which can further be simplified into

$$\vartheta_1 \varrho_1 = \vartheta_0 \varrho_0; \text{ for } n = 0, \quad (4.12)$$

and

$$\vartheta_{n+1} \varrho_{n+1} = \vartheta_n \varrho_n + \sum_{j=1}^n \varphi_j \varrho_{n+1-j} + \beta_{n+1} \varrho_0; \text{ for } n \geq 1, \quad (4.13)$$

where

$$\vartheta_n = 4a_n \cos \beta h + b_n, \quad (4.14)$$

and

$$\begin{aligned} \vartheta_{n+1} &= (a_{n+1} + c_{n+1})(\cos \beta h - i \sin \beta h) + b_{n+1} \\ &= 4a_{n+1} \cos \beta h + b_{n+1}, \text{ (since } a_n = c_n), \end{aligned} \quad (4.15)$$

for all $n = 0, 1, 2, \dots, N-1$.

Now (4.12) and (4.13), respectively, yields

$$\varrho_1 = \frac{\vartheta_0}{\vartheta_1} \varrho_0; \text{ for } n = 0, \quad (4.16)$$

and

$$\varrho_{n+1} = \frac{\vartheta_n}{\vartheta_{n+1}} \varrho_n + \frac{1}{\vartheta_{n+1}} \sum_{j=1}^n \varphi_j \varrho_{n+1-j} + \frac{\beta_{n+1}}{\vartheta_{n+1}} \varrho_0; \text{ for } n \geq 1. \quad (4.17)$$

We note, $\square$

Remark 4.2. The ratio $\left| \frac{\vartheta_n}{\vartheta_{n+1}} \right|$ is monotonically decreasing for all $n = 0, 1, 2, \dots, N-1$, i.e.,

$$\begin{aligned} \left| \frac{\vartheta_n}{\vartheta_{n+1}} \right| &= \left| \frac{4a_n \cos \beta h + b_n}{4a_{n+1} \cos \beta h + b_{n+1}} \right| \\ &\leq \left| \frac{4a_n \cos \beta h + b_n}{4a_n \cos \beta h + b_n} \right| = 1. \end{aligned} \quad (4.18)$$

We can also establish

Proposition 4.3. Suppose ϱ_{n+1} is a solution to (4.16) and (4.17), then $|\varrho_{n+1}| \leq |\varrho_0|$; for all $n = 0, 1, 2, \dots, N-1$.

Proof. Suppose $n = 0$, then in view of Remark 4.2, we have from (4.16):

$$|\varrho_1| = \left| \frac{\vartheta_0}{\vartheta_1} \varrho_0 \right| \leq |\varrho_0|. \quad (4.19)$$

Now let $n \geq 1$ and suppose that $|\varrho_n| \leq |\varrho_0|$ for all n , then from (4.17), we have

$$|\varrho_{n+1}| = \left| \frac{\vartheta_n}{\vartheta_{n+1}} \varrho_n + \frac{1}{\vartheta_{n+1}} \sum_{j=1}^n \varphi_j \varrho_{n+1-j} + \frac{\beta_{n+1}}{\vartheta_{n+1}} \varrho_0 \right|$$

$$\begin{aligned}
&\leq \left| \frac{\vartheta_n}{\vartheta_{n+1}} Q_n \right| + \left| \frac{1}{\vartheta_{n+1}} \sum_{j=1}^n \varphi_j Q_{n+1-j} \right| + \left| \frac{\beta_{n+1}}{\vartheta_{n+1}} Q_0 \right|, \\
&\leq \left| \frac{\vartheta_n}{\vartheta_{n+1}} \right| |Q_n| + \left| \frac{1}{\vartheta_{n+1}} \right| \left| \sum_{j=1}^n \varphi_j Q_{n+1-j} \right| + \left| \frac{\beta_{n+1}}{\vartheta_{n+1}} \right| |Q_0|, \\
&\leq |Q_n| + \left| \sum_{j=1}^n \varphi_j Q_{n+1-j} \right| + |\beta_{n+1} Q_0|, \\
&= |Q_n| + \varphi_1 |Q_n| + \varphi_2 |Q_{n-1}| + \cdots + \varphi_n |Q_1| + \beta_{n+1} |Q_0|, \\
&\leq |Q_0| + \varphi_1 |Q_0| + \varphi_2 |Q_0| + \cdots + \varphi_n |Q_0| + \beta_{n+1} |Q_0|, \\
&= \left(\sum_{j=0}^n \varphi_j + \beta_{n+1} \right) |Q_0| = (1 - \beta_{n+1} + \beta_{n+1}) |Q_0|, \text{ (see Remark 3.1),} \\
&= |Q_0|.
\end{aligned} \tag{4.20}$$

The above along with (4.8) yields $\|\epsilon^n\|_2 \leq \|\epsilon^0\|_2$ for all $n = 1, 2, \dots, N-1$ which completes the proof of Theorem 4.1. $\square$

4.2. Convergence analysis of the numerical method

Theorem 4.4. The difference scheme (3.10)–(3.11) along with (2.14) is convergent and has the order $\mathcal{O}(k^2 + h^2)$.

Proof. Let U_l^n be the exact solution to (3.1) and (2.14) at the grid point (S_l, t_n) . Define $e_l^n = U_l^n - V_l^n$, such that $e_0^n = e_L^n = 0$ for all $n = 0, 1, 2, \dots, N-1$.

Define the truncation error by

$$\begin{aligned}
R_l^n &= \sum_{j=0}^n \beta_j [V(S_l, t_{n+1-j}) - V(S_l, t_{n-j})] - \mu V(S_l, t_{n+1}) - \omega [V(S_{l+1}, t_{n+1}) - V(S_l, t_{n+1})] \\
&\quad + P_\alpha [V(S_{l+1}, t_{n+1}) - 2V(S_l, t_{n+1}) + V(S_{l-1}, t_{n+1}) + V(S_{l+1}, t_n) \\
&\quad - 2V(S_l, t_n) + V(S_{l-1}, t_n)],
\end{aligned} \tag{4.21}$$

where

$$\mu = k^\alpha r t^{1-\alpha}, \quad \omega = k^\alpha (r - D) l t^{1-\alpha},$$

and

$$P_\alpha = \frac{k^\alpha \Gamma(2 - \alpha) (l^2 \sigma^2 \Gamma(1 + \alpha))}{2}.$$

Note that R_l^n in (4.21) is obtained from (3.9) by multiplying both sides by the term $k^\alpha \Gamma(2 - \alpha)$.

To proceed further, we need the following two propositions: $\square$

Proposition 4.5. There exist a constant $\tilde{C}$ such that $|R_l^n| \leq \tilde{C}(k^2 + h^2)$ for all $l = 0, 1, 2, \dots, L-1$ and $n = 0, 1, 2, \dots, N$.

Proof. Define

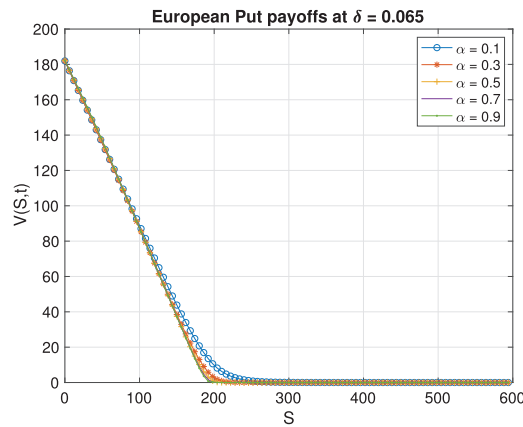
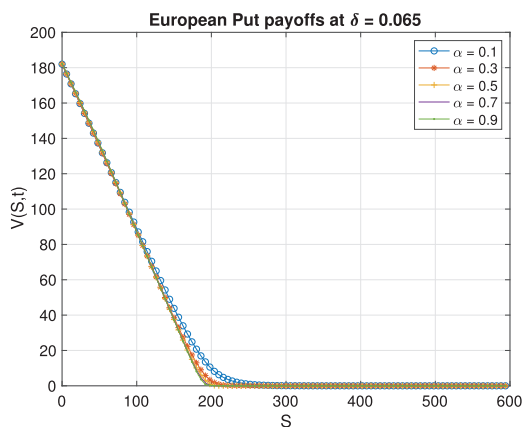
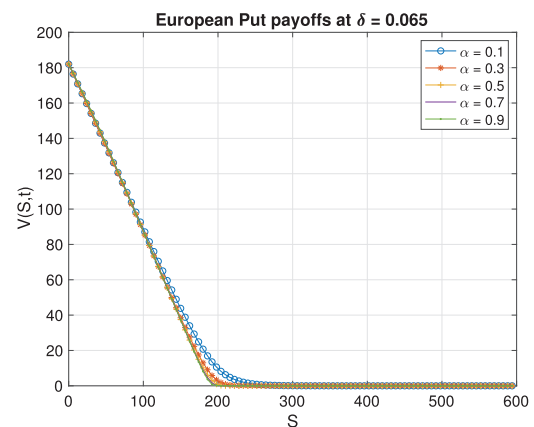
$$L_k^\alpha V(S_l, t_{n+1}) := \frac{k^{-2\alpha}}{2\Gamma(2 - \alpha)} \sum_{j=0}^{n+1} \beta_j [V(S_l, t_{n+1-j}) - V(S_l, t_{n-j})],$$

then

$$\begin{aligned}
&\left| \frac{\partial^\alpha V(S_l, t_{n+1})}{\partial t^\alpha} - L_k^\alpha V(S_l, t_{n+1}) \right| \\
&\leq \frac{1}{2\Gamma(1 - \alpha)} \sum_{j=0}^{n+1} \int_{jk}^{(j+1)k} \left| \frac{\partial V(S_l, \tau)}{\partial \tau} - \frac{(V(S_l, t_{n+1-j}) - V(S_l, t_{n-j}))}{k^2} \right| \frac{d\tau}{(t_{n+1} - \tau)^\alpha}, \\
&\leq \frac{1}{2\Gamma(1 - \alpha)} k^2 \sum_{j=0}^{n+1} \int_{jk}^{(j+1)k} \frac{d\tau}{(t_{n+1} - \tau)^\alpha}, \\
&\leq \frac{C}{2\Gamma(1 - \alpha)} k^2 \int_{jk}^{(j+1)k} \frac{d\tau}{(t_{n+1} - \tau)^\alpha}, \\
&\leq C_1 k^2,
\end{aligned} \tag{4.22}$$

where C and C_1 are constants independent of h and k . Therefore

$$\frac{k^{-\alpha}}{2\Gamma(2 - \alpha)} \sum_{j=0}^{n+1} \beta_j [V(S_l, t_{n-j+2}) - V(S_l, t_{n-j+1})] = \frac{\partial^\alpha V(S_l, t_{n+1})}{\partial t^\alpha} + C_1 k^2. \tag{4.23}$$

(a) $\delta = 0.025$ (b) $\delta = 0.055$ (c) $\delta = 0.065$ **Fig. 1.** Example 5.1 maturity payoffs ($t = T$) for $\alpha = (0.1, 0.3, 0.5, 0.7, 0.9)$

In a similar way, we can show that

$$\frac{V(S_{l+1}, t_{n+1}) - V(S_{l-1}, t_{n+1})}{2h} = \frac{\partial V(S_l, t_{n+1})}{\partial S} + C_2 h^2, \quad (4.24)$$

and

$$\begin{aligned} \frac{V_{l+1}^{n+1} - 2V_l^{n+1} + V_{l-1}^{n+1} + V_{l+1}^n - 2V_l^n + V_{l-1}^n}{2h^2} \\ = \frac{\partial^2 V(S_l, t_{n+1})}{\partial S^2} + C_3 h^2. \end{aligned} \quad (4.25)$$

Substituting (4.23) - (4.25) into (4.21) and simplifying, we obtain

$$\begin{aligned} R_l^n = k^\alpha \Gamma(2 - \alpha) \left[\frac{\partial^\alpha V(S_l, t_n)}{\partial t^\alpha} - \frac{\mu}{k^\alpha \Gamma(2 - \alpha)} V(S_l, t_n) \right. \\ \left. - \frac{\omega}{k^\alpha \Gamma(2 - \alpha)} \frac{\partial V(S_l, t_n)}{\partial S} + P_\alpha \frac{\partial^2 V(S_l, t_n)}{\partial S^2} \right] \\ + C_1 k^{2+\alpha} + C_2 k^\alpha h^2 + C_3 k^\alpha h^2. \end{aligned} \quad (4.26)$$

This implies that

$$\begin{aligned} |R_l^n| &\leq C_4 (k^{2+\alpha} + k^\alpha (h^2 + h^2)), \\ &\leq \hat{C} (k^{2+\alpha} + k^\alpha h^2), \\ &= \hat{C} k^\alpha (k^2 + h^2), \\ &\leq \tilde{C} (k^2 + h^2), \quad (\text{since } 0 < k^\alpha < 1). \end{aligned} \quad (4.27)$$

Since the errors e_l^n satisfy equation (3.10)-(3.11), substituting e_l^n into (3.10)-(3.11) we obtain

$$a_1 e_{l-1}^1 + b_1 e_l^1 + c_1 e_{l+1}^1 = R_l^1, \quad (4.28)$$

and

$$\begin{aligned} a_{n+1} e_{l-1}^{n+1} + b_{n+1} e_l^{n+1} + c_{n+1} e_{l+1}^{n+1} = a_n e_{l-1}^n + b_n e_l^n + c_n e_{l+1}^n \\ + \sum_{j=1}^n \varphi_j e_l^{n+1-j} + R_l^{n+1}. \end{aligned} \quad (4.29)$$

Let us define the following grid functions

$$e^n(S) = \begin{cases} e_l^n, & \text{when } S_l - \frac{h}{2} < S \leq S_l + \frac{h}{2}, \quad l = 1, 2, \dots, L-1, \\ 0, & \text{when } 0 \leq S < \frac{h}{2} \text{ or } S_{\max} - \frac{h}{2} < S \leq S_{\max} + \frac{h}{2}, \end{cases} \quad (4.30)$$

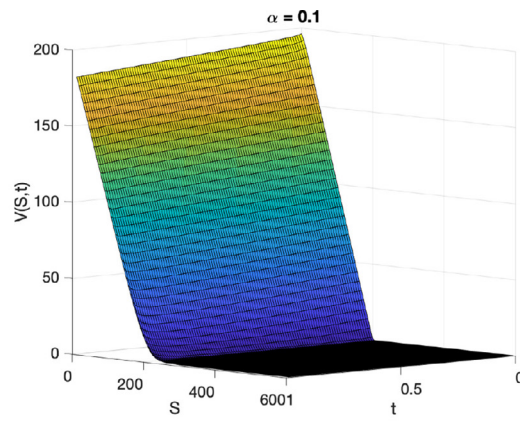
and

$$R^n(S) = \begin{cases} R_l^n, & \text{when } S_l - \frac{h}{2} < S \leq S_l + \frac{h}{2}, \quad l = 1, 2, \dots, L-1, \\ 0, & \text{when } 0 \leq S < \frac{h}{2} \text{ or } S_{\max} - \frac{h}{2} < S \leq S_{\max} + \frac{h}{2}, \end{cases} \quad (4.31)$$

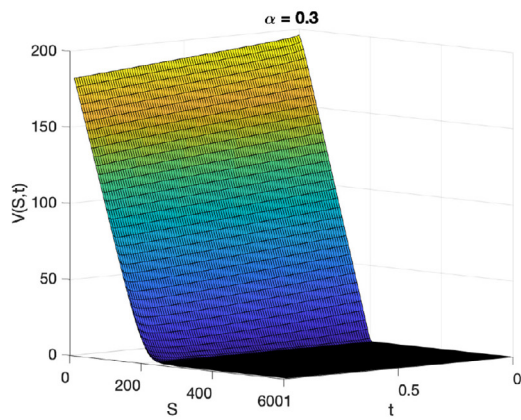
respectively. Note that, the functions $e^n(S)$ and R_l^n can be expanded into the following Fourier series representations:

$$e^n(S) = \sum_{j=1}^{\infty} \lambda_n(j) e^{i2\pi jS/S_{\max}}, \quad n = 0, 1, 2, \dots, N, \quad (4.32)$$

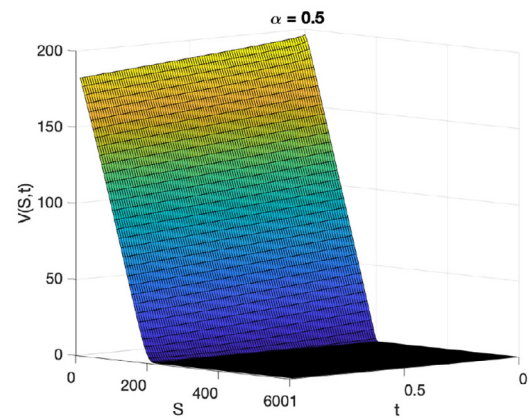
$$R^n(S) = \sum_{j=1}^{\infty} \eta_n(j) e^{i2\pi jS/S_{\max}}, \quad n = 0, 1, 2, \dots, N, \quad (4.33)$$



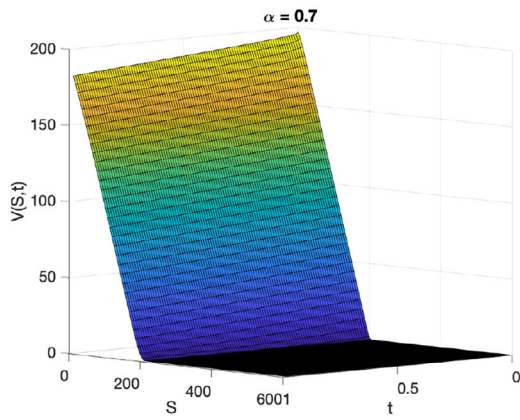
(a) Payoff for $\alpha = 0.1$



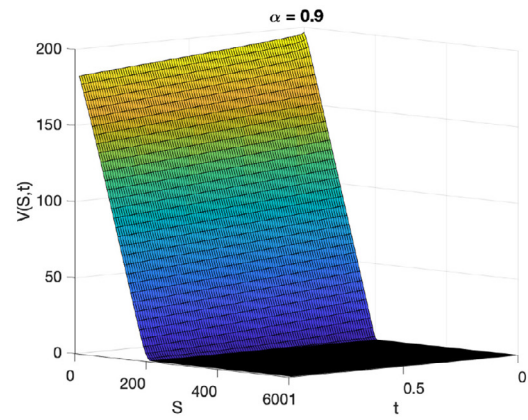
(b) Payoff for $\alpha = 0.3$



(c) Payoff for $\alpha = 0.5$



(d) Payoff for $\alpha = 0.7$



(e) Payoff for $\alpha = 0.9$

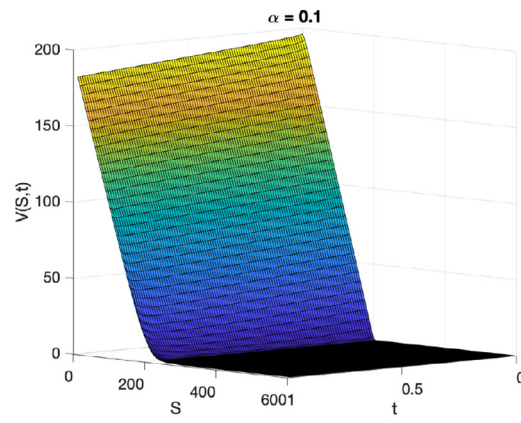
Fig. 2. Example 5.1 general payoffs for $\alpha = (0.1, 0.3, 0.5, 0.7, 0.9)$ and $\delta = 0.025$.

for

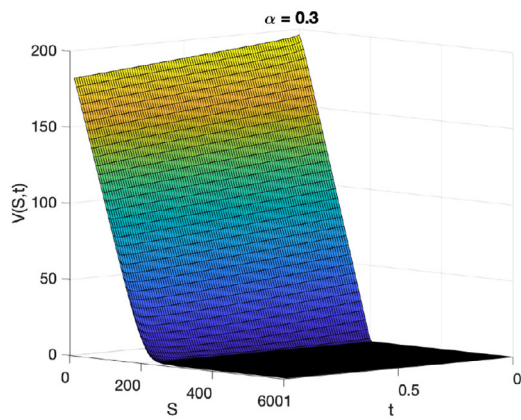
and

$$\lambda_n(j) = \frac{1}{S_{\max}} \int_0^{S_{\max}} e^n(S) e^{-i2\pi jS/S_{\max}} dS; \quad n = 0, 1, 2, \dots, N, \quad (4.34)$$

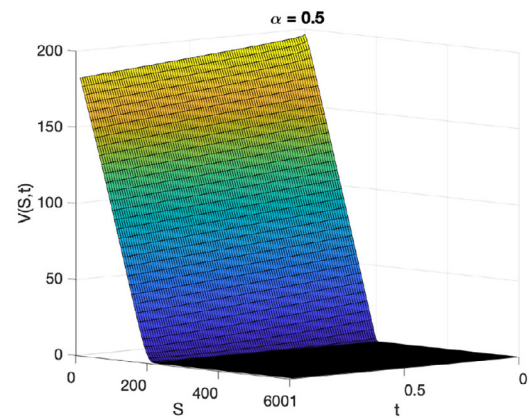
$$\eta_n(j) = \frac{1}{S_{\max}} \int_0^{S_{\max}} R^n(S) e^{-i2\pi jS/S_{\max}} dS; \quad n = 0, 1, 2, \dots, N. \quad (4.35)$$



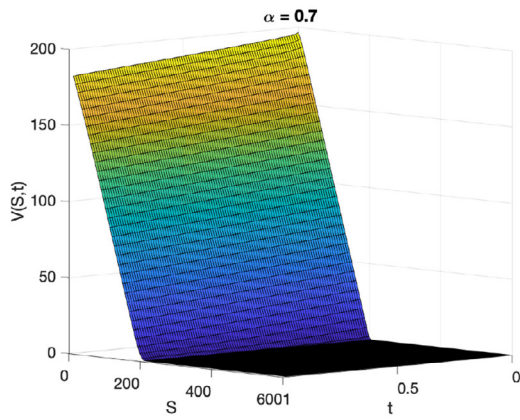
(a) Payoff for $\alpha = 0.1$



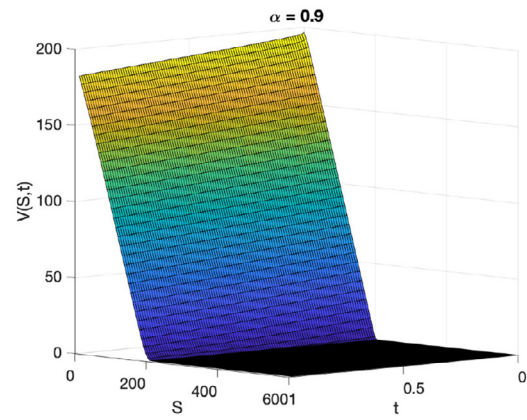
(b) Payoff for $\alpha = 0.3$



(c) Payoff for $\alpha = 0.5$



(d) Payoff for $\alpha = 0.7$



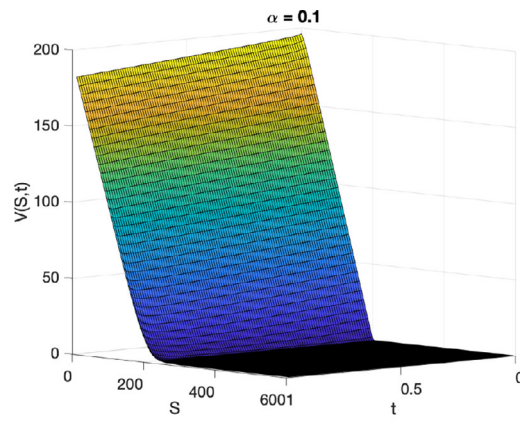
(e) Payoff for $\alpha = 0.9$

Fig. 3. Example 5.1 general payoffs for $\alpha = (0.1, 0.3, 0.5, 0.7, 0.9)$ and $\delta = 0.055$.

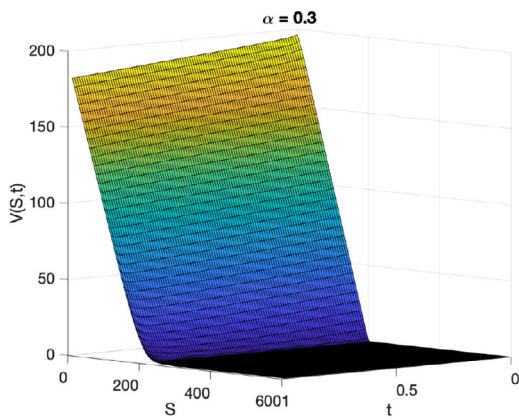
Let $e^n = (e_1^n, e_2^n, \dots, e_{L-1}^n)^T$ and $R^n = (R_1^n, R_2^n, \dots, R_{L-1}^n)^T$ with their second norms as

$$\|e^n\|_2 = \left(\sum_{l=1}^{L-1} h |e_l^n|^2 \right)^{1/2} = \left(\int_0^{S_{\max}} |e^n(S)|^2 dS \right)^{1/2};$$

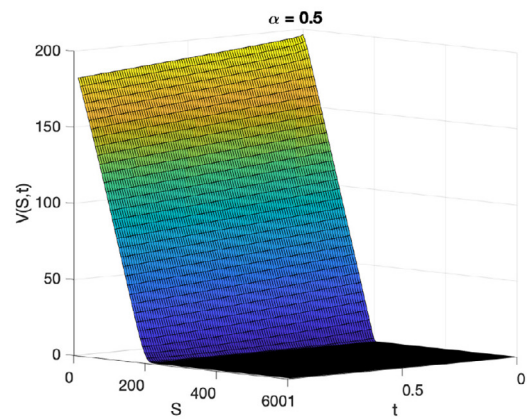
$$n = 0, 1, 2, \dots, N$$
(4.36)



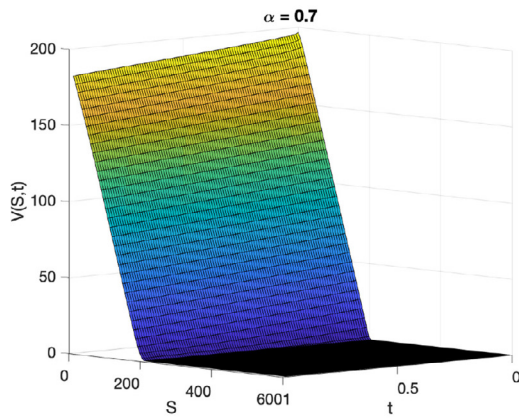
(a) Payoff for $\alpha = 0.1$



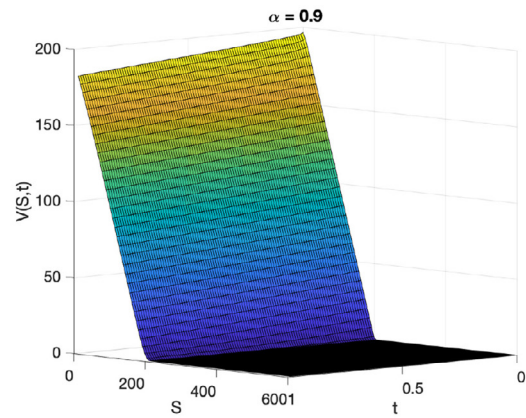
(b) Payoff for $\alpha = 0.3$



(c) Payoff for $\alpha = 0.5$



(d) Payoff for $\alpha = 0.7$



(e) Payoff for $\alpha = 0.9$

Fig. 4. Example 5.1 general payoffs for $\alpha = (0.1, 0.3, 0.5, 0.7, 0.9)$ and $\delta = 0.065$.

and

$$\|R^n\|_2 = \left(\sum_{l=1}^{L-1} h |R_l^n|^2 \right)^{1/2} = \left(\int_0^{S_{\max}} |R^n(S)|^2 dS \right)^{1/2};$$

$$n = 1, 2, 3, \dots, N.$$

(4.37)

Now the Parseval equality imply

$$\int_0^{S_{\max}} |e^n(S)|^2 dS = \sum_{j=-\infty}^{\infty} |\lambda_n(j)|^2, \quad (4.38)$$

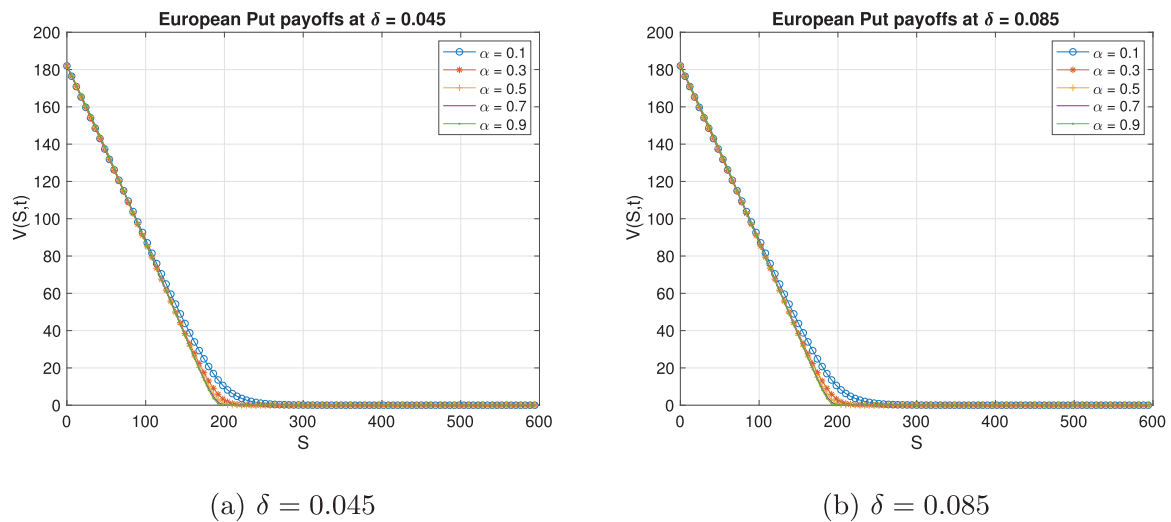


Fig. 5. Example 5.2 maturity payoffs ($t = T$) for $\alpha = (0.1, 0.3, 0.5, 0.7, 0.9)$

and

$$\int_0^{S_{\max}} |R^n(S)|^2 dS = \sum_{j=-\infty}^{\infty} |\eta_n(j)|^2, \quad (4.39)$$

which leads to

$$\|e^n\|_2^2 = \int_0^{S_{\max}} |e^n(S)|^2 dS = \sum_{j=-\infty}^{\infty} |\lambda_n(j)|^2; \quad n = 0, 1, 2, \dots, N, \quad (4.40)$$

and

$$\|R^n\|_2^2 = \int_0^{S_{\max}} |R^n(S)|^2 dS = \sum_{j=-\infty}^{\infty} |\eta_n(j)|^2; \quad n = 1, 2, 3, \dots, N. \quad (4.41)$$

Suppose that

$$e^n = \lambda_n e^{i\beta h}, \quad (4.42)$$

and

$$R^n = \eta_n e^{i\beta h}, \quad (4.43)$$

for $\beta = 2\pi j/S_{\max}$ and $i = \sqrt{-1}$. Substituting e^n and R^n into (4.28) and (4.29) and further simplifying, we obtain

$$\lambda_1 \vartheta_1 = \eta_1, \text{ when } n = 0, \quad (4.44)$$

and

$$\lambda_{n+1} \vartheta_{n+1} = \vartheta_n \lambda_n + \sum_{j=1}^n \varphi_j \lambda_{n-j+1} + \eta_{n+1}, \quad (4.45)$$

where $\vartheta_{n+1} = 4c_{n+1} \cos \beta h + b_{n+1}$, for $n = 1, 2, \dots, N$. $\square$

Proposition 4.6. If λ_n is a solution to (4.44) and (4.45) for all $n = 0, 1, 2, \dots, N$, then $|\lambda_n| \leq C |\eta_1|$.

Proof. From (4.27) coupled with (4.37), we get

$$\|R^{n+1}\|_2 \leq \tilde{C}(k^2 + h^2); \quad n = 0, 1, 2, \dots, N.$$

Therefore, from (4.41), we have

$$|\eta_{n+1}| = |\eta_{n+1}(j)| \leq C |\eta_1| = C |\eta_1(j)|; \quad n = 0, 1, 2, \dots, N, \quad (4.46)$$

for some constant C , independent of h and k .

Let $n = 0$, then, from (4.44) we have

$$|\lambda_1| = \left| \frac{1}{\vartheta_1} \eta_1 \right| \leq C |\eta_1|. \quad (4.47)$$

Now suppose $|\lambda_n| \leq C |\eta_1|$ for $n \geq 1$. Then we show that it is also true for $n + 1$. To see this, we have from (4.45)

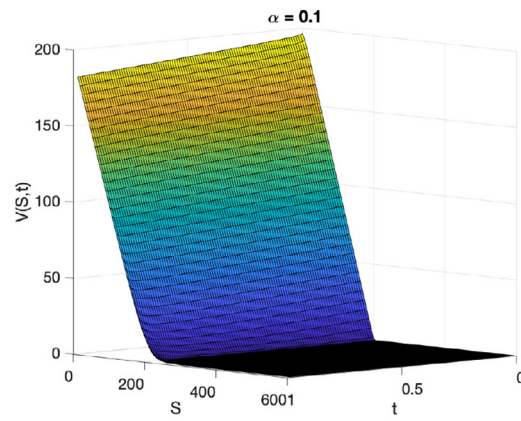
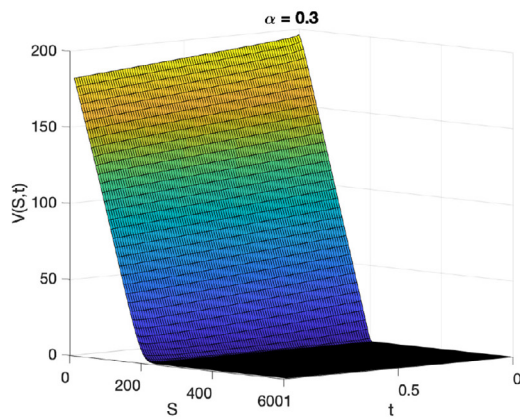
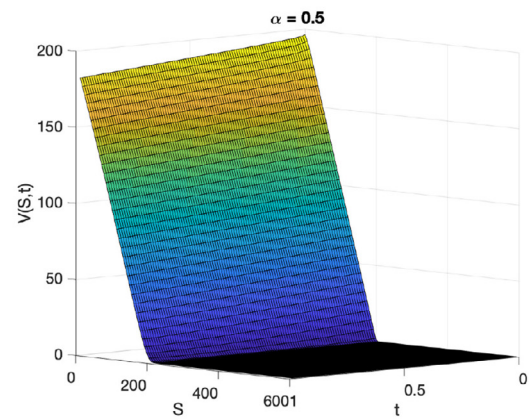
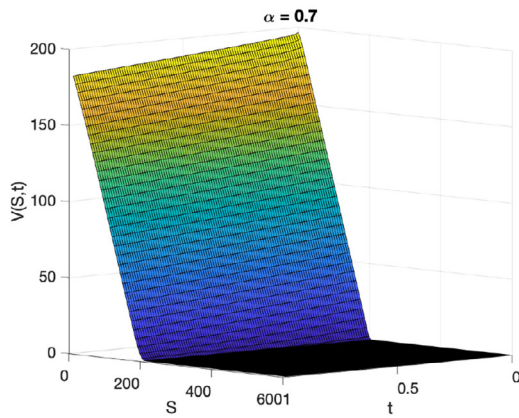
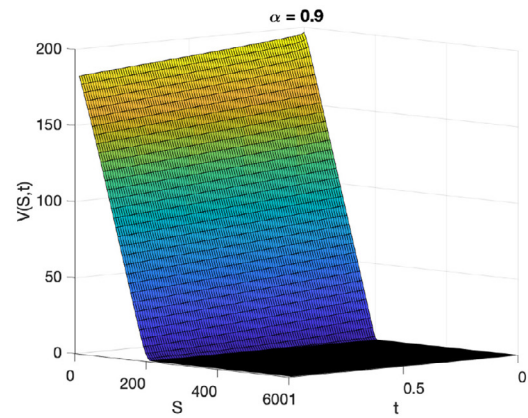
$$\begin{aligned} |\lambda_{n+1}| &= \left| \frac{1}{\vartheta_{n+1}} \left(\vartheta_n \lambda_n + \sum_{j=1}^n \varphi_j \lambda_{n-j+1} + \eta_{n+1} \right) \right|, \\ &\leq \frac{1}{|\vartheta_{n+1}|} \left(|\vartheta_n| |\lambda_n| + \sum_{j=1}^n |\varphi_j| |\lambda_{n-j+1}| + |\eta_{n+1}| \right), \\ &\leq \frac{1}{|\vartheta_{n+1}|} \left(C_0 |\eta_1| + \sum_{j=1}^n \varphi_j C_j |\eta_1| + C_{n+1} |\eta_1| \right), \\ &\leq \frac{1}{|\vartheta_{n+1}|} \left(\bar{C} |\eta_1| + \sum_{j=1}^n \varphi_j \bar{C} |\eta_1| + \bar{C} |\eta_1| \right), \\ &\quad \left(\bar{C} = \max_{0 \leq j \leq n+1} \{C_j\} \right), \\ &\leq \hat{C} |\eta_1| + \sum_{j=1}^n \varphi_j \hat{C} |\eta_1| + \hat{C} |\eta_1|, \quad \left(\hat{C} = \frac{\bar{C}}{|\vartheta_{n+1}|} \right), \\ &= \hat{C} \left\{ |\eta_1| + \sum_{j=1}^n \varphi_j |\eta_1| + |\eta_1| \right\}, \\ &= \hat{C} (3 - \beta_{n+1}) |\eta_1|, \\ &= C |\eta_1|. \end{aligned} \quad (4.48)$$

Combining the results obtained above, we see that the scheme (3.10)-(3.11) is convergent, which completes the proof of Theorem 4.4. $\square$

5. Numerical results and discussions

In this section, we present two numerical examples on the pricing of European put options through the proposed approach.

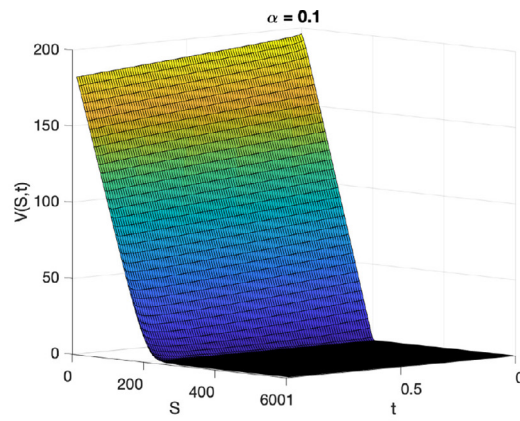
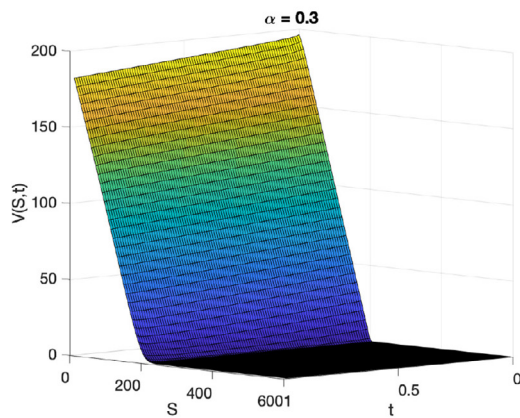
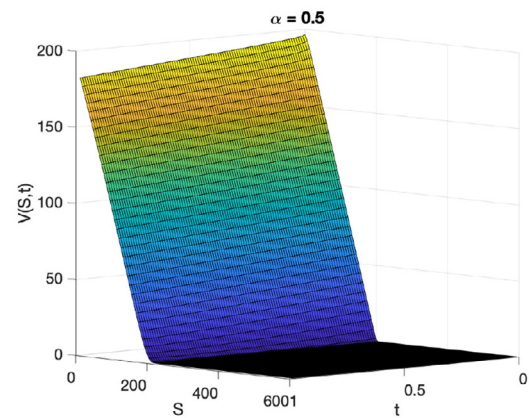
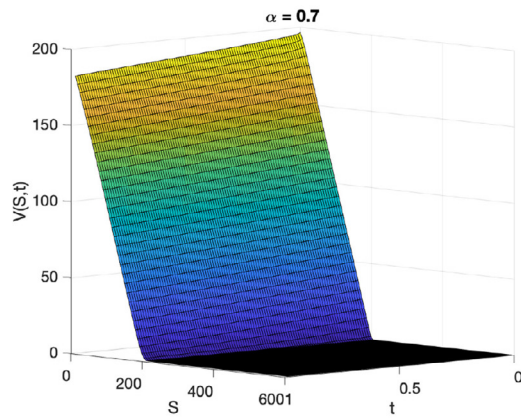
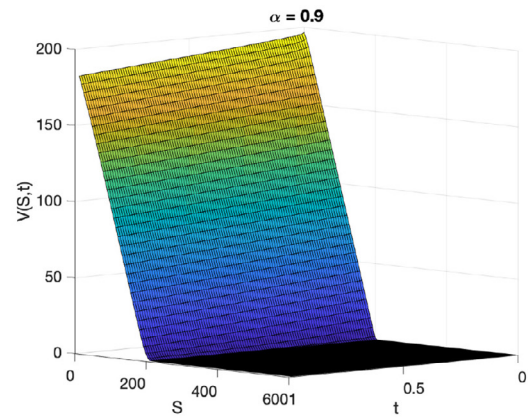
Example 5.1. Consider the tfBS equation (2.13) subject to initial and boundary conditions (2.14) for pricing a standard European put option with the following parameters: $K = 150$, $r = 0.055$, $\sigma = 0.01$, $T = 1$, $S_{\max} = 450$, $L = 100$, $N = 100$, $\delta = 0.025, 0.055$ and 0.065 .

(a) Payoff for $\alpha = 0.1$ (b) Payoff for $\alpha = 0.3$ (c) Payoff for $\alpha = 0.5$ (d) Payoff for $\alpha = 0.7$ (e) Payoff for $\alpha = 0.9$ **Fig. 6.** Example 5.2 general payoffs for $\alpha = (0.1, 0.3, 0.5, 0.7, 0.9)$ and $\delta = 0.085$.

To check for the effects of the change in some key option parameters on the efficiency of the model as well as the numerical method herein, we consider a second example with two different set of dividend yields δ , a different risk-free interest rate r , a dif-

ferent measure of volatility σ , and a larger value for $S_{\max}$ under the same set of α values as in Example 5.1.

Example 5.2. Consider the *tfBS* equation (2.13) subject to initial and boundary conditions (2.14) for pricing a standard European

(a) Payoff for $\alpha = 0.1$ (b) Payoff for $\alpha = 0.3$ (c) Payoff for $\alpha = 0.5$ (d) Payoff for $\alpha = 0.7$ (e) Payoff for $\alpha = 0.9$ **Fig. 7.** Example 5.2 General payoffs for $\alpha = (0.1, 0.3, 0.5, 0.7, 0.9)$ and $\delta = 0.085$.

put option with the following parameters: $K = 200$, $r = 0.065$, $\sigma = 0.025$, $T = 1$, $S_{\max} = 600$, $L = 100$, $N = 100$, $\delta = 0.045$ and 0.085 .

The results for the above two examples are presented in Tables 1,2,3,4. The European option maturity payoff curves obtained from model (2.13) under market parameters specified in

example 5.1 and example 5.2 are presented in fig. 1 and fig. 5, respectively. To see how the payoffs generally behave throughout the life-time of the option contract at the considered values α , results are presented in figs. 2,3,4,6 and 7.

Table 1
Maximum absolute errors for [example 5.1](#) with $r = 0.055$ and $\delta = 0.025$.

α	$N = 100$	$N = 200$	$N = 400$	$N = 800$	$N = 1600$
0.1	1.3714e-02	3.4691e-03	8.7754e-04	2.2198e-04	5.5953e-05
0.2	1.2394e-02	3.1352e-03	7.9308e-04	2.0062e-04	5.0548e-05
0.3	1.0162e-02	2.5506e-03	6.5025e-04	1.6449e-04	4.1409e-05
0.4	9.5623e-03	2.4089e-03	6.1188e-04	1.5478e-04	3.7153e-05
0.5	8.5515e-03	2.0632e-03	5.4720e-04	1.3842e-04	3.4915e-05
0.6	8.0315e-03	2.0216e-03	5.1393e-04	1.3000e-04	3.2785e-05
0.7	7.5609e-03	1.9126e-03	4.8381e-04	1.2239e-04	3.0759e-05
0.8	7.3065e-03	1.8183e-03	4.6753e-04	1.1827e-04	2.9717e-05
0.9	7.1436e-03	1.8170e-03	4.5711e-04	1.1563e-04	2.9250e-05
1.0	7.1202e-03	1.7911e-03	4.5561e-04	1.1525e-04	2.9154e-05

Table 2
Convergence rates for [example 5.1](#) with $r = 0.055$ and $\delta = 0.025$.

α	$N = 200$	$N = 400$	$N = 800$	$N = 1600$
0.1	1.91	1.95	1.98	1.99
0.2	1.92	1.96	1.98	1.99
0.3	1.93	1.96	1.98	1.99
0.4	1.93	1.96	1.98	1.99
0.5	1.93	1.97	1.98	1.99
0.6	1.94	1.97	1.98	1.99
0.7	1.94	1.97	1.98	1.99
0.8	1.94	1.97	1.98	1.99
0.9	1.94	1.97	1.98	1.99
1.0	1.94	1.97	1.98	1.99

Table 3
Maximum absolute errors for [example 5.2](#) with $r = 0.065$ and $\delta = 0.085$.

α	$N = 100$	$N = 200$	$N = 400$	$N = 800$	$N = 1600$
0.1	6.5592e-02	1.6592e-02	4.1972e-03	1.0617e-03	2.6857e-04
0.2	5.8088e-03	1.4694e-02	3.7170e-03	9.4025e-04	2.3784e-04
0.3	5.2147e-02	1.3191e-02	3.3368e-03	8.4408e-04	2.1352e-04
0.4	4.7443e-02	1.2001e-02	3.0358e-03	7.6794e-04	1.9426e-04
0.5	4.3746e-02	1.1066e-02	2.7993e-03	7.0810e-04	1.7912e-04
0.6	4.0893e-02	1.0344e-02	2.6167e-03	6.6192e-04	1.6544e-04
0.7	3.8773e-02	9.8080e-03	2.4810e-03	6.2760e-04	1.5676e-04
0.8	3.7318e-02	9.4400e-03	2.3879e-03	6.0405e-04	1.5080e-04
0.9	3.6499e-02	9.2328e-03	2.3355e-03	5.9079e-04	1.4745e-04
1.0	3.6328e-02	9.1895e-03	2.3246e-03	5.8803e-04	1.4670e-04

Table 4
Convergence rates for [example 5.2](#) with $r = 0.065$ and $\delta = 0.085$.

α	$N = 200$	$N = 400$	$N = 800$	$N = 1600$
0.1	1.95	1.98	1.99	1.99
0.2	1.96	1.98	1.99	1.99
0.3	1.96	1.98	1.99	1.99
0.4	1.96	1.98	1.99	2.00
0.5	1.97	1.98	1.99	2.00
0.6	1.97	1.98	1.99	2.00
0.7	1.97	1.98	1.99	2.00
0.8	1.97	1.98	1.99	2.00
0.9	1.97	1.98	1.99	2.01
1.0	1.97	1.98	1.99	2.01

Overall, numerical results herein confirm theoretical deductions made in [Section 4](#) that the proposed scheme is unconditionally stable and converges with order $\mathcal{O}(h^2, k^2)$. These can be seen from [tables 3](#) and [4](#) for market parameters specified under [example 5.1](#) and [tables 3](#) and [4](#) for market parameters specified in [example 5.2](#). Furthermore, the convergence results presented in [tables 1](#) and [3](#) show that the numerical method for calculating European stock option premiums is robust regardless of the value of α . Also, from the numerical point of view, since the proposed method is unconditionally stable, the choice of α does not affect the overall convergence of the method. These observations

are validated by the results presented in [tables 1](#) and [3](#). Results in [tables 2](#) and [4](#) also confirm that the method is stable irrespective of the value of α . Therefore, α can be chosen to be small or large and the method will still maintain its' overall numerical convergence and stability properties, hence showing the applications of our proposed method in solving a larger class of problems.

From the application point of view, one observe that the general pay off curves are more smooth when $1/2 \leq \alpha < 1$, and otherwise when $0 < \alpha < 1/2$. The latter is signalling asymmetric performance of the fractional approach as compared to the classical approach. This is so because, when $1/2 \leq \alpha < 1$, the underlying fractional stochastic process is more persistent characterised by positively correlated increments with memory. It is therefore expected that, compared to the standard Gaussian distribution in classical Black-Scholes models, the fractional stochastic process with $1/2 \leq \alpha < 1$ is more effective. The case for $0 < \alpha < 1/2$ represent an anti-persistent regime of the underlying fractional stochastic process. Under this regime, the underlying fractional stochastic process is characterised by negatively correlated stock price increments, hence its under performance when compared to the case when $1/2 \leq \alpha < 1$. However, the restriction on $1/2 \leq \alpha < 1$ is not too large a hindrance because, normal markets's conditions would not require extreme lower values of α . That is so because, under extremely lower values of α , the stock price returns are negatively correlated which in turn violates the fundamental principle of no arbitrage in asset pricing theory. Furthermore, contrasting the obtained option premium curves at different dividend yields, regardless of the value of α , we observe that, high dividends, yield high put option premiums. This is true because, if the underlying stock pays dividends, then its price is expected to relatively drop by the value of the dividend payout. Therefore, changing the dividend yield does have a significant effect on the put option premium results.

6. Concluding remarks and scope for future direction

In this paper, we proposed a robust numerical scheme for solving tfBS-PDE in pricing continuous dividend paying European stock option. Two numerical examples are presented. Theoretical and numerical results herein indicate that the fractional Black-Scholes approach is a very efficient option pricing tool compared to its classical counterpart. In addition, the proposed numerical method has also proven to be robust and efficient in solving tfBS-PDEs. A worth noting observation is that, general option payoff curves and profiles are more smooth when $1/2 \leq \alpha < 1$, and otherwise when $0 < \alpha < 1/2$. With the latter signalling asymmetric performance of the approach and the proposed method herein. Overall, these imply that, the approach is more robust when $1/2 \leq \alpha < 1$ compared to when $0 < \alpha < 1/2$. These is so because, it is expected that when $1/2 \leq \alpha < 1$, the proposed fractional stochastic process is persistently characterised by positive correlated stock price increments, hence, a better performance is attained when $1/2 \leq \alpha < 1$ compared to when $0 < \alpha < 1/2$. In conclusion therefore, our results confirm theoretical observations and general consensus in literature that, stock market dynamics are of a power law nature, and follows heavy tailed distributions with hereditary properties.

For further research, the exploration of other higher order numerical methods as well as extending some of the newly formulated results obtained using recent methods such as the q -homotopy analysis transform method to financial problems remain a subject of future research.

Declaration of Competing Interest

No [Eqn 2.2](#) and [2.3](#) interests to declare.

CRediT authorship contribution statement

Samuel M Nuugulu: Conceptualization, Investigation, Methodology, Software, Writing - original draft. **Frednard Gideon:** Supervision, Validation. **Kailash C Patidar:** Conceptualization, Supervision, Validation.

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